Nigmat Rahim

+86 13521755312 | nigmatrahim4@gmail.com
GitHub: github.com/Nigmat-future
Research Interests: Bioinformatics & Biocomputing

EDUCATION

Peking University Aug 2022 - Jul 2026
B.S. in Medical Technology (Laboratory) Institute of Medical Technology Beijing
GPA: 3.14/4.00 (rank 16/35); Junior year GPA: 3.65/4.00 (rank 6/33)
Core courses: Internal Medicine (94.9); Clinical Microbiology (91.2); Pattern Recognition & Artificial Intelligence (89.0); Clinical Biochemistry (88.7); Radiotherapy Practicum (88.6); Clinical Molecular Biology (86.9); Pathophysiology (86.0); Instrumental Analysis (85.5); Clinical Hematology (85.2)

HONORS & AWARDS

Outstanding Research Award, Peking University Health Science Center 2025
Stanford RNA 3D Folding Challenge (Kaggle) | Bronze Medal (Top 9%) Ranked 131st out of 1,516 global teams 2025
Second Prize, Undergraduate Innovation Program, Peking University Third Hospital, 2024 2024

PUBLICATION & MANUSCRIPTS

[1] ***Rahim, N.†***, Chen, R.†, Yan, C.†, Wu, Z., Wu, C., Lin, Y., Zhang, H., Yu, Z., Tong, J., Jing, H., & Zhang, W.* "HubGPT: Retrieval-Augmented Identification of Hub Genes and Key Pathways." Under review at Advance science. (*Co-first Author*)*

[2] Qu, Y.†, Wei, Q.†, Wu, C.†, ***Rahim, N.†***, Luo, J.†, Deng, J.†, Hu, W., Liu, J., Wang, L., Lin, X., Xiao, J., Yu, Z., Yan, C., Liu, X., Yuan, X., Tong, J., Wang, J., Zhu, M., Yang, P., Chen, Y., Gao, S., Zhang, H., Zhou, H., Tan, Q., Wan, W., Jing, H., & Zhang, W.* "Comprehensive Multi-omics and Single-Cell Analysis Reveals Ferroptosis Signature Associated with Tumor Microenvironment Remodeling and Therapeutic Decision-Making across Pan-Cancer." International Journal of Surgery. Accepted. (*Co-first Author*)*

[3] Zhang, C.†, Wei, Q.†, ***Rahim, N.†***, Bian, G.†, Wu, C., Wang, J., Yang, P., Hua, J., Zhu, M., Chen, Y., Liu, Y., Gao, S., Hou, B., Zhang, W., & Jing, H. "CD73 Inhibitor Significantly Enhances the Antitumor Activity of Selinexor in Multiple Myeloma by Specifically Activating CD8+ T Cells." Submitted to Hemasphere. (*Co-first Author*)

[4] ***Rahim, N***. "Bispecific CAR-T vs. BCMA-Specific CAR-T: Advancing Treatment for Relapsed/Refractory Multiple Myeloma." Accepted at FENBC 2025 (Conference). (*First Author*)

RESEARCH EXPERIENCE

ThyroPADA: An Agentic AI System for Enhancing Cellular-Level Region of Interest Detection in Thyroid FNA Biopsies Jan 2025 - Present
Peking University Third Hospital | Mentor: Dr. Weilong Zhang

- Architected an AI-agent diagnostic system integrating RAG and multimodal LLMs (GPT-4o/4.1) to autonomously extract and quantify 16 cytopathological features from thyroid whole-slide images (WSIs), replacing manual feature annotation by pathologists.
- Curated the largest multi-center thyroid cytopathology dataset to date (7 centers, 12,000+ cases) and trained an optimized GBM model via grid search across 7 ML algorithms, achieving an AUC of 0.96 on the internal test set.
- Validated model robustness across 6 independent external centers (AUC: 0.935–0.974), demonstrating consistent generalizability across diverse clinical settings and patient populations.
- Conducted a human-AI comparison study showing that AI-assisted diagnosis (Acc: 90.7%) outperformed both senior and junior pathologists working independently, supporting the clinical utility of the system.

HubGPT: retrieval-augmented identification of hub genes and key pathways Nov 2024 - Jan 2026
Peking University Third Hospital | Mentor: Dr. Weilong Zhang

- Created HubGPT, a novel computational framework using Retrieval-Augmented Generation (RAG) to prioritize key genes and pathways in cancer, addressing the high false-positive rates of traditional KEGG enrichment analysis.
- Defined and implemented custom scoring metrics (Pri-value and R-value) to quantify pathway essentiality, successfully distinguishing context-specific oncogenic drivers from housekeeping genes across 24 cancer types.
- Applied the model to filter biological noise, successfully removing irrelevant pathways (e.g., removing Malaria/Measles pathways from Bladder Cancer enrichment results) while elevating canonical drivers like p53 and PI3K-Akt.
- Validated the model's ability to identify druggable targets; the system autonomously reconstructed the AML signaling hierarchy, ranking clinically approved targets (e.g., FLT3, KIT, STAT5B) as top-priority nodes.
- Correlated AI-derived importance scores with functional dependency data from DepMap.

Comprehensive Multi-omics and Single-Cell Analysis Reveals Ferroptosis Signature Associated with Tumor Microenvironment Remodeling and Therapeutic Decision-Making across Pan-Cancer Feb 2024 - Dec 2025

Peking University Third Hospital | Mentor: Dr. Weilong Zhang

- Developed a prognostic ferroptosis signature (Fp.Sig) by integrating multi-omics data from 10,510 TCGA samples across 33 cancer types using 10 distinct machine learning algorithms (e.g., RSF, Enet), achieving a C-index up to 0.67.
- Engineered and deployed an interactive web application using R Shiny for clinicians to predict patient survival rates and immunotherapy response based on the Fp.Sig model.
- Experimentally validated the Fp.Sig model, demonstrating a significant negative correlation with SN-38 drug sensitivity in 5 myeloma cell lines (R^2=0.86, p=0.005 in GDSC1 cohort).

CD73 inhibitor significantly enhances the antitumor activity of selinexor by specifically activating CD8 T cells Oct 2023 - Oct 2025

Peking University Third Hospital | Mentor: Dr. Weilong Zhang

- Analyzed single-cell RNA-seq data from a murine tumor model (J558/BALB/c) to elucidate the synergistic antitumor mechanism, identifying significant downregulation of the Malignant_Enpp1 tumor-promoting subpopulation.
- Reconstructed the differentiation trajectories of over 45,000 T/NK cells using Monocle, revealing that the combination therapy specifically activates and enhances the cytotoxic function of CD8+ T cells (upregulation of IFN-γ, Granzyme B, Prf1).

Bispecific CAR-T vs. BCMA-Specific CAR-T: Advancing Treatment for Relapsed/Refractory Multiple Myeloma

Shanghai Jiao Tong University School of Medicine, Ruijin Hospital | Mentor: Prof. Dakang Xu Jan 2025 - Apr 2025

- Led a comprehensive analysis of 7 clinical trials (n=190), demonstrating the superior efficacy of bispecific CAR-T over monospecific CAR-T in R/R multiple myeloma (ORR: 89.6% vs 73%; median PFS: ~18 months vs 8.8 months).
- Investigated drug resistance mechanisms by analyzing scRNA-seq data, identifying the MIF/CD74 pathway as a key contributor to immune escape in BCMA-negative resistant patients ($p < 0.001$).

INTERNSHIP EXPERIENCE

Peking University Third Hospital (Beijing Third Hospital) May 2025 - Nov 2025

Clinical Laboratory

- Completed a 6-month clinical rotation across 5 subspecialties (biochemistry, immunology, hematology, microbiology, molecular diagnostics). Gained hands-on experience in qPCR/real-time PCR, ELISA, flow-related immunoassays, and internal quality control (IQC) procedures.

COMMUNITY EXPERIENCE

Peking University Health Science Center Sep 2023 - Present

Study committee, Vice Monitor

- Led an R-language study group with graduate mentors; covered bioinformatics basics including TCGA/GEO usage and deployment of AlphaFold 2/3.
- Organized reading groups on Transformer and ViT frameworks to deepen understanding of DL advances and applications in bioinformatics.

SKILLS LIST

Programming & Data Science

- Advanced: Python (Pandas, NumPy, PyTorch), R (tidyverse, Seurat, ggplot2, Shiny)
- Proficient: Scikit-learn, Machine Learning (RSF, Elastic Net), Shell Scripting, Retrieval-Augmented Generation (RAG) Implementation

Bioinformatics & Computational Biology

- Analysis: Single-Cell RNA-seq, Multi-omics Integration, Gene Set Enrichment Analysis (GSEA), Pathway Analysis (KEGG/GO), Survival Analysis
- Visualization: BioRender, Illustrator

Languages

- Bilingual: Uyghur, Mandarin Chinese
- Fluent: English (TOEFL: 99)， CET-4/6
- Basic: French, German